Xiaozhu Zhang

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Research interests

- Model selection, uncertainty quantification
- Causal discovery, multi-source learning
- Mixed-integer programming

Education

University of Washington, Seattle — Seattle, WA
Ph.D. in **Statistics**, GPA: 3.95/4.00 — 09/2023 – 06/2028
Advisor: Armeen Taeb

Duke University — Durham, NC
M.S. in **Statistical Science**, GPA: 4.00/4.00 — 08/2021 – 05/2023
Advisor: Yuansi Chen

Hunan University — Changsha, China
B.S. in **Statistics**, GPA: 3.94/4.00 — 09/2016 – 07/2020

Papers

- **Zhang, X.**, Mazumder, R., and Taeb, A. (2026). Convex mixed-integer programming for invariant predictions from multi-environment data. *arXiv preprint arXiv:2610.38942.*.
- **Zhang, X.**, Keret, N., Taeb, A., and Shojaie A. (2025). Convex mixed-integer programming for causal additive models with optimization and statistical guarantees. *arXiv preprint arXiv:2511.21126.* ***Major revision at JASA***.
- **Zhang, X.**, Bien, J., and Taeb, A. (2025). Quantifying uncertainty and stability among highly correlated predictors: a subspace perspective. *arXiv preprint arXiv:2505.06760. In submission.*
- **Zhang, X.** (2023). *Stable Variable Selection for Sparse Linear Regression in a Non-uniqueness Regime.* [Master's thesis, Duke University]. DukeSpace.
- **Zhang, X.**, and Reiter, J. P. (2022). A Generator for Generalized Inverse Gaussian Distributions. *arXiv preprint arXiv:2211.13049.*

Talks

- Short talk, *FIM-IMS Joint Workshop on Mathematics of Data Science*, November 2026, Singapore
- Contributed talk, *Joint Statistical Meetings*, August 2026, Boston
- Invited talk, *IMS Annual Meeting*, July 2026, Salzburg
- Invited talk, *IMS Asia Pacific Rim Meeting*, June 2026, Hong Kong
- Poster, *American Causal Inference Conference*, May 2026, Salt Lake City
- Contributed talk, *the 5th Biennial Meeting of the Pacific Northwest Section of SIAM*, October 2025, Seattle
- Invited talk, *Joint Conference on Statistics and Data Science in China*, July 2025, Hangzhou

Awards

- Graduate Student Conference Presentation Awards, University of Washington, 2026
- Dean's Research Award for Master's Students, Duke University, 2022
- Outstanding Undergraduates, Hunan Province, 2020
- China National Scholarship, 2017 – 2019
- National 1st Prize & Best Paper Award in China Undergraduate Mathematical Contest in Modeling, 2018

Teaching

Teaching assistant, University of Washington

- STAT 583: Advanced Theory of Statistical Inference III (Spring 2026)
- STAT 528: Applied Statistics Capstone (Winter 2025, Winter 2026)
- STAT 435: Introduction to Statistical Machine Learning (Spring 2025)
- STAT 396: Finite Markov Chains and Monte-Carlo Methods (Spring 2024)
- STAT 390: Statistical Methods in Engineering and Science (Autumn 2023)
- STAT 311: Elements of Statistical Methods (Autumn 2026)

Teaching assistant, Duke University

- STA 602: Bayesian Statistical Modeling (Spring 2023)
- STA 523: Programming for Statistical Science (Fall 2022)